Review of ECO3121

1 Dummy Variables

- If Y_i is binary, then $E(Y_i|X_i) = \Pr(Y_i = 1|X_i)$
- Three models:

  I. linear probability model (linear multiple regression)

  $P(y = 1|x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$

  Interpretation:

  β_1 describe the effect of the independent variables on the probability that $y = 1$, holding other variables constant.

  Disadvantages:

  i. Predicted probabilities may be larger than one or smaller than zero

  ii. The estimated partial effects are constant throughout all the possible values of independent variables

  II. probit(cumulative standard normal distribution)

  $\Pr(Y_i = 1) = \Phi(Z)$

  with $Z = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{\text{ki}}$

  Interpretation:

  e.g. x_1 increases from 0.1 to 0.2:

  $\widehat{\Delta \Pr(Y_i = 1)} = \Phi(\beta_0 + \beta_1 * 0.2 + \cdots + \beta_k x_{\text{ki}}) - \Phi(\beta_0 + \beta_1 * 0.1 + \cdots + \beta_k x_{\text{ki}})$

  III. logit(cumulative standard logistic distribution)

  $\Pr(Y_i = 1) = \frac{1}{1 + e^{-z}}$

  with $Z = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{\text{ki}}$

- Probit and logit are estimated via maximum likelihood

2 Instrumental Variables

- Motivation ($E(u|X) \neq 0$)

  I. Omitted variables

  II. Measurement error (X is measured with error)

  Attenuation Bias

  III. Simultaneity (X causes Y, Y causes X)

- A valid instrument Z must satisfy two conditions:

  I. Relevance: $\text{corr}(Z, X) \neq 0$

II. Exogeneity: $\text{corr}(Z, u) = 0$

- Two Satge Least Squares (2SLS)

  I. First, isolate the part of X_j that is uncorrelated with u

  $$X_{\text{ij}} = \pi_{j,0} + \pi_{j,1} Z_{1i} + \cdots + \pi_{j,m} Z_{\text{mi}} + \pi_{j,m+1} W_{1i} + \cdots + \pi_{j,m+r} W_{\text{ri}} + v_{\text{ji}} \quad \text{for } j = 1, \ldots, k$$

  II. Replace X_{ij} by $\widehat{X_{\text{ij}}}$ in the regression of interest

  $$Y_i = \beta_0 + \beta_1 \widehat{X_{1i}} + \ldots + \beta_k \widehat{X_{\text{ki}}} + \beta_{k+1} W_{1i} + \cdots + \beta_{k+r} W_{\text{ri}} + u_i$$

- Check Instrument Validity

  I. Relevance

  $$X_{\text{ij}} = \pi_{j,0} + \pi_{j,1} Z_{1i} + \cdots + \pi_{j,m} Z_{\text{mi}} + \pi_{j,m+1} W_{1i} + \cdots + \pi_{j,m+r} W_{\text{ri}} + v_{\text{ji}}$$

  H_0: $\pi_1 = \ldots = \pi_m = 0$

  Rule of thumb: $F > 10$. If $F < 10,$ you may have weak IV issue.

  II. Exogeneity

  The test is J-test(Overidentification test), which is constructed using the 2SLS residuals. $J = \text{mF}$

  H_0: All the m instruments are exogenous $(m > k)$

  If the J-test is rejected, then at least some of your instruments are endogenous.

3 Heteroskedasticity

- Consequences of Heteroskadasticity for OLS

  I. OLS still unbiased and consistent, and interpretation of R-squared is not changed.

  II. Heteroskedasticity invalidates variance formulas for OLS estimators, and the usual F-tests and t-tests are not valid. Besides, OLS is no longer BLUE.

- Heteroskedasticity-Robust Inference after OLS Estimation

- Testing for Heteroskedasticity

  I. Breusch-Pagan test

  Regress squared residuals on all explanatory varaibles and compute the usual F test.

  II. White test

  Regress squared residuals on all explanatory variables, their squares, and interactions.

  Alternative form: Regress squared residuals on the predicted value of y and its square.

- Weighted least squares estimation(known heteroskedasticity function)

  $\text{Var}(u_i | X_i) = \sigma^2 h(X_i), h(X_i) = h_i > 0$

  $y_i^* = \frac{y_i}{\sqrt{h_i}}$, $x_i^* = \frac{x_i}{\sqrt{h_i}}$

- Feasible Generalized Least Squares(unknown heteroskedasticity function)

  Assumed general form: $\text{Var}(u|X) = \sigma^2 \exp(\delta_0 + \delta_1 x_1 + \cdots + \delta_k x_k) = \sigma^2 h(X)$

4 Panal Data Analysis

- Panel data(longitudianl data): The behavior of entities are observed across time.

  Balance panel: no missing observations

  Unbalanced panel: some entities are not obeserved for some time periods.

- $Y_{\text{it}} = \beta_0 + \beta_1 X_{\text{it}} + \gamma Z_i + \delta S_t + e_{\text{it}}, i = 1, \ldots, n, t = 1, \ldots, T$

  Z_i: Vary across entities but do not vary over time

  S_t: Vary over time but not across entities

  Estimation:

  I. n-1 and T-1 binary regressor OLS regression

  II. Entity and time-demeaned OLS regression

  III. Changes OLS regression

- Difference in Differences

  I. Estimand: $\tau_{\text{ATT}} = E[Y_{1i}(1) - Y_{0i}(1)|D = 1]$

	Post-Period(T=1)	Pre-Period(T=0)
Treated $D_i = 1$	$E[Y_{1i}(1)\|D_i = 1]$	$E[Y_{0i}(0)\|D_i = 1]$
Controled $D_i = 0$	$E[Y_{0i}(1)\|D_i = 0]$	$E[Y_{0i}(0)\|D_i = 0]$

  Use: $\{E[Y_i(1)|D_i = 1]\text{-}E[Y_i(1)|D_i = 0]\} - \{E[Y_i(0)|D_i = 1]\text{-}E[Y_i(0)|D_i = 0]\}$

  Assume(Parallel trends): $E[Y_{0i}(1) - Y_{0i}(0)|D_i = 1] = E[Y_{0i}(1) - Y_{0i}(0)|D_i = 0]$

  It requires that the trend in the outcome variable for both treatment and control groups during the pre-treatment era are similar

  II. $\boldsymbol{y_{\text{it}} = \beta_0 + \beta_1 I(\text{treat}_{\text{it}}) + \beta_2 I(\text{post}_{\text{it}}) + \beta_3 I(\text{treat}_{\text{it}}) * I(\text{post}_{\text{it}}) + \varepsilon_{\text{it}}}$

	Post	Pre	Difference
Treatment	$\beta_0 + \beta_1 + \beta_2 + \beta_3$	$\beta_0 + \beta_1$	$\beta_2 + \beta_3$
Control	$\beta_0 + \beta_2$	β_0	β_2
Difference	$\beta_1 + \beta_3$	β_1	β_3